

Suman Mandal

github/man-su-97 | sumanmandal15135@gmail.com | linkedin/suman-mandal | +91 8145020621

SUMMARY

Full-stack developer in GenAI and RAG-based systems — shipping LLM-powered products end to end, collaborating with product and engineering teams to deliver FastAPI services, pgvector pipelines, and Next.js frontends for multi-tenant SaaS powered by LLMs.

EXPERIENCE

A&J INTELLI SYSTEMS PVT LTD | FULL STACK DEVELOPER

Aug 2025 – Present | Gurugram (Remote)

- Led design and delivery of a RAG pipeline — data ingestion, chunking, embeddings, pgvector semantic search, and prompt assembly — powering a natural language chat interface over structured datasets via DuckDB.
- Built Next.js + React Query dashboard (drag-and-drop layout, KPI cards, auto-generated chart recommendations) and FastAPI / NestJS APIs handling DuckDB queries and LLM responses.
- Drove production delivery — PostgreSQL RLS tenant isolation, Stripe subscription billing, Redis rate limiting, JWT auth bridge, and Docker Compose deployment.

NASHBUD INC | FULL STACK ENGINEER

Mar 2024 – Jul 2025 | Kolkata

- Shipped MERN-stack eCommerce — product catalog (search, filter, sort, pagination), cart with dynamic pricing, multi-step checkout, and order tracking across 3 fulfillment states; grew user engagement by 25%.
- Integrated Razorpay with HMAC-SHA256 verification, idempotency guards, and a circuit breaker (trips after 3 failures, self-heals on cooldown); built admin dashboard with 10+ analytics charts; increased conversion rates by 15%.
- Implemented Firebase Auth (RBAC, 404-cloaked routes); migrated MongoDB → PostgreSQL 16 via Drizzle ORM with row-level locking.

UPLIANCE.AI | FRONTEND INTERN

Nov 2023 – Jan 2024 | Bengaluru

- Rewrote device control logic with Redux Toolkit + Thunk; applied observer pattern in JavaScript for efficient state handling in CookClub.

PROJECTS

PRISM | NEXT.JS, FASTAPI, POSTGRES SQL, DUCKDB, ANTHROPIC API, DOCKER

github.com/man-su-97/Prism

- Multi-tenant SaaS: users upload CSV / Excel / Sheets and get auto-generated dashboards (KPI tiles, line/bar/pie charts) with ReactJS drag-and-drop layout; DuckDB + Parquet analytics engine with PostgreSQL RLS per-org data isolation.
- LLM agent in a structured tool-use loop — generates and validates SQL, blocks prompt-injection via `validate_sql`, and streams responses over SSE with multi-turn prompt caching.
- Stripe billing (Free / Pro / Team) with monthly LLM token entitlements, Redis hourly rate limiting, and revocable share links with email delivery.

NEWS AGGREGATOR API | PERSONAL PROJECT

GITHUB.COM/MAN-SU-97/NEWS-APP-SERVER

- Async FastAPI service ingesting RSS feeds and REST APIs into a unified canonical schema; strict layered architecture with PostgreSQL deduplication (`ON CONFLICT DO NOTHING`) for fully idempotent multi-source ingestion.
- SQLAlchemy 2.0 + asyncpg, Alembic migrations, and FastAPI `Depends` injection for clean testability and session-scoped transactions.

SKILLS

LANGUAGES

Python • JavaScript • TypeScript • SQL

FRONTEND

ReactJS • Next.js • Tailwind CSS • React Query • Redux Toolkit • Zustand

BACKEND

Node.js • Express.js • FastAPI • NestJS • REST APIs • PostgreSQL

AI & GENAI

RAG • pgvector • LangChain • LangGraph • LLM APIs (Claude / OpenAI) • Prompt Engineering • Embeddings • DuckDB

TOOLS & PLATFORMS

Git • Docker • GitHub Actions • AWS • Azure • Redis • Vercel

EDUCATION

BENGAL INSTITUTE OF TECHNOLOGY

B.TECH IN INFORMATION TECHNOLOGY | CGPA: 9.11/10

2019 – 2023 | Kolkata, West Bengal